IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-526

1. Topic (12 words max)

E-Commerce Platform for Personalized Retail Experiences

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In the evolving landscape of online commerce, personalization has become a critical factor in improving user experience and increasing sales. Traditional e-commerce platforms often offer generic interfaces and static promotions, failing to consider individual customer behaviors, seasonal trends, or local language preferences. Additionally, small to medium retailers lack access to intelligent tools that help optimize their sales strategies, monitor inventory effectively, and engage customers in a context-aware and personalized manner.

The core challenge addressed by this research is the absence of a unified, AI-powered e-commerce platform that integrates behavioral analytics, voice-based interaction in local languages, and explainable AI-based decision support for shop owners. Current platforms do not offer localized voice support in Sinhala, which limits accessibility for non-English speaking users, especially in rural and semi-urban areas in Sri Lanka.

Furthermore, while many systems provide basic reporting dashboards, few integrate XAI (Explainable AI) to help decision-makers understand why certain recommendations—like restocking or promotion targeting—are made. This transparency is key to improving trust and adoption.

This research project aims to fill these gaps by creating a comprehensive, AI-integrated e-commerce platform tailored for both shop owners and customers. It leverages unsupervised learning to segment customer behavior, provides real-time inventory and revenue insights through explainable analytics, and enhances user interaction via a Sinhala-based voice assistant.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a full-stack e-commerce platform that provides tailored user experiences and shop-owner decision support, powered by AI technologies.

On the customer side, users interact with a bilingual interface (Sinhala/English), browse products with real-time availability and pricing, and receive AI-generated promotional recommendations based on their behavior and past purchases. The system also integrates a Sinhala-based voice assistant trained on shopping-specific datasets, allowing customers to ask product-related questions naturally.

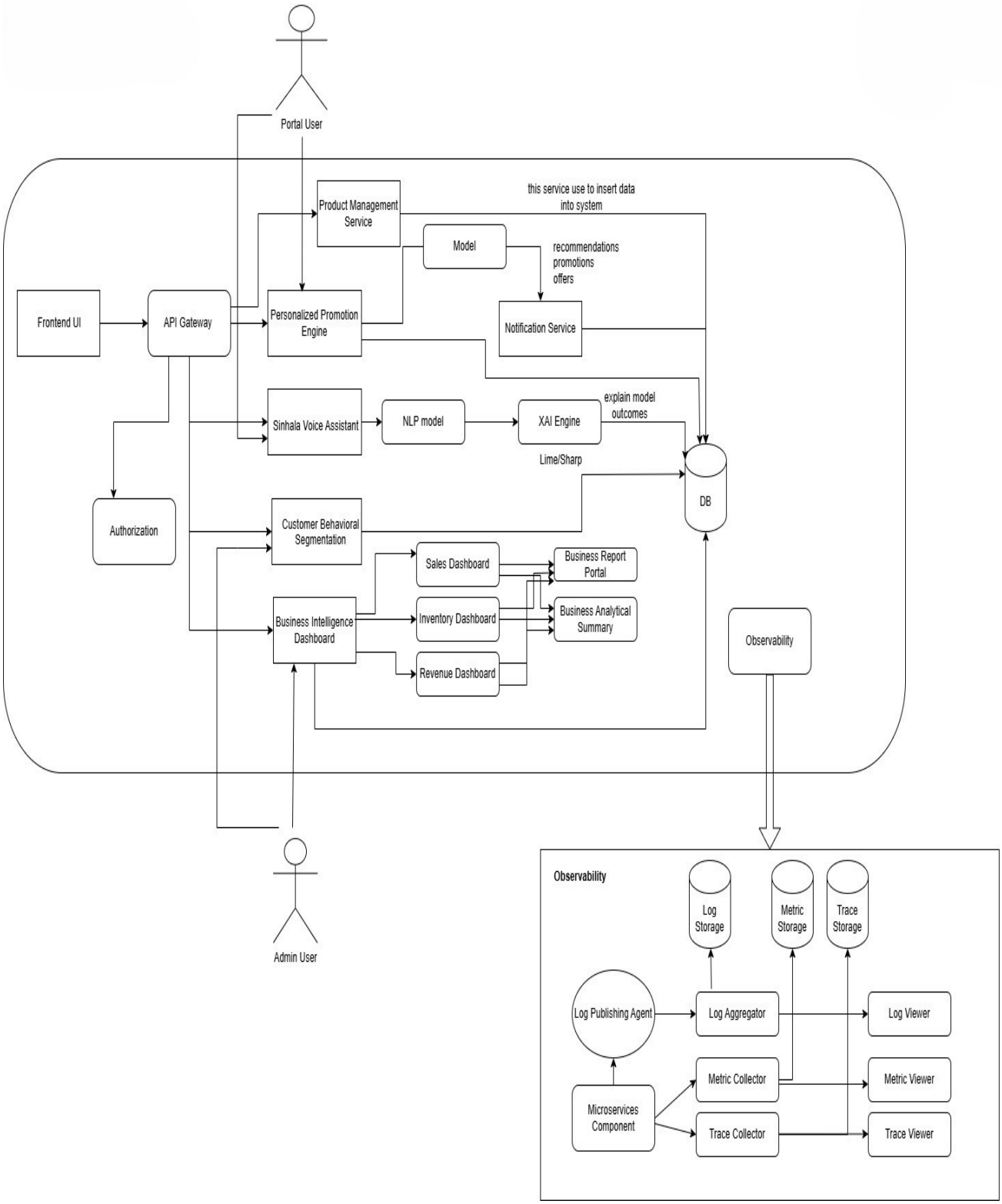
On the retailer side, shop owners access a powerful analytical dashboard featuring real-time sales data, inventory forecasting, and loyalty impact analysis. The dashboard uses explainable AI models to justify restocking alerts and promotional campaign outcomes, enhancing trust in automated recommendations. The system also segments customers using clustering algorithms (e.g., K-Means), enabling micro-targeted loyalty programs.

The backend leverages machine learning models for behavioral pattern detection and recommendation generation, while the frontend offers a responsive interface designed for accessibility across devices.

The conceptual architecture includes:

- Frontend UI: Web/mobile interface for customers and retailers
- ML Engine: Behavior analysis, promotions, segmentation
- Voice NLP Engine: Sinhala speech recognition and intent detection
- Analytics Module: BI dashboard with XAI integration
- Database Layer: Product, user, and transaction records
- API Layer: Communication between components

Smart Ecommerce Platform for personolized retail experiences



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project spans multiple specialized domains:

Artificial Intelligence & Machine Learning: For customer segmentation (unsupervised learning), recommendation systems, promotion optimization, and Explainable AI (XAI) for inventory and revenue analysis.

Natural Language Processing (NLP): Specialized in Sinhala language processing, speech-to-text, and voice intent recognition for creating the Sinhala voice assistant.

Software Engineering: Building scalable full-stack web applications using modern frameworks and deploying machine learning pipelines in production environments.

Business Intelligence (BI): Creating dashboards that visualize sales trends, inventory metrics, and loyalty analytics to guide business decisions.

Data Requirements:

- Historical Sales Data: To train the recommendation and forecasting models.
- Product Catalogs & Inventory Data: For price tracking, stock alerts, and restocking logic.
- Voice Data in Sinhala: To train and fine-tune NLP models for voice interaction.
- Customer Behavior Logs: Including purchase frequency, categories, seasonal habits.

Open-source datasets and synthetic data may initially be used for model training. Real-world anonymized data may be integrated via pilot testing phases in partnership with small retailers for validation.

8. Objectives and Novelty

Main Objective: To design and develop an AI-powered e-commerce platform that delivers personalized retail experiences through behavior-based promotions, Sinhala voice assistance, and explainable analytics dashboards for both customers and shop owners.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Pramuditha W. G. S. IT22588814	Design and develop a Sinhala-based voice assistant for e-commerce interaction.	• Collect and process Sinhala-language voice samples. • Train NLP models to recognize and interpret customer shopping queries. • Implement voice-to-text and intent detection pipelines. • Integrate assistant with the recommendation system.	Sinhala NLP integration for contextual e-commerce interactions: a novel feature in local online platforms
Nayanapriya H. T. E. IT22367044	Develop a personalized AI-based promotion engine.	• Analyze customer purchase history and preferences. • Build machine learning models to recommend products promotions. • Generate dynamic promotions (seasonal, bundled, or preference-based) according to users interaction history of the products. • Suggest individual user based promotions according to users interaction patterns. • Integrate the promotion engine with the user interface.	AI-generated real-time **personalized** promotions with contextual adaptation to customer behavior.(Instead of bulk promotions suggests user based promotions by analyzing user buying patterns for long time)

Nimsara S. P. N. IT22326072	Implement customer behavioral segmentation and targeted loyalty programs.	• Collect and preprocess customer transaction data. • Apply unsupervised learning (e.g., K-Means) to identify shopping patterns. • Group customers based on frequency, value, and product preferences. • Design personalized loyalty tiers and dynamic discounting strategies.	Micro-targeted loyalty programs using machine-learned customer clusters instead of static segmentation.
Dinuwan W. T. IT22306968	Build a Business Intelligence (BI) dashboard with explainable AI (XAI) for shop owners.	A. Data Processing • Collect and clean sales/inventory data. • Aggregate metrics (e.g., revenue by customer segment). B. AI Integration • Implement demand forecasting. • Add **Explainable AI (XAI)** to justify insights (e.g., "Restock advised due to 20% demand surge") C. Dashboard Development • Build interactive charts D. Validation • Compare AI predictions vs. actual sales data. • Test with mock users (e.g., shop owners).	Explainable AI-powered analytics for inventory and loyalty—enabling transparent and data-driven decisions.

8. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Pramuditha W. G. S. IT22588814	Handles natural language processing (NLP) components and UI integration for Sinhala voice support, demonstrating applied AI and frontend specialization. In addition to core development, containerizes the voice assistant and frontend services, tests them locally, and pushes versioned Docker images to Amazon ECR for integration into the deployment pipeline.
Nayanapriya H. T. E. IT22367044	Focuses on machine learning integration and full-stack development to create the AI-based promotion engine, aligning with core SE skills in backend logic, recommendation systems, and API development. In addition, setup the Kubernetes environment using Amazon EKS, writes deployment YAML files, configures services and ingress, and ensures smooth deployment using kubectl
Nimsara S. P. N. IT22326072	Works on customer behavioral clustering and loyalty program integration, utilizing algorithm design and model training with direct application in software architecture. Alongside this, implements Jenkins CI/CD pipelines to automate image building and deployment processes, pushing services to Amazon ECR and deploying them to the EKS cluster with GitHub webhook integration.
Dinuwan W. T. IT22306968	Leads the BI dashboard and explainable AI modules, focusing on data engineering, visualization, and transparency in AI—key components of modern software engineering solutions. In addition to development, validates the deployed system on EKS, applies monitoring tools such as Postman and K9s, and documents the complete deployment process, including scaling strategies and troubleshooting steps.

9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Prof.	Dilshan	De Silva	
Co-Supervisor	Ms.	Chamali	Pabasara	27.06.2025 (on behalf of Ms. Chamali)
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.